

KayosCrypto – Executive Summary PQC (24 Nov 2025)

1. Snapshot for Decision-Makers

- **Status**: PQC validation stack fully integrated in docs, diagnostics, and MPC-N (audit trail). 9/9 core tests + PractRand PQC suite green.
- **Differential**: Geometric Fishbone architecture (Fibonacci → Ezekiel → Core) now paired with Kyber/Dilithium benchmarks and ChaCha20-whitened entropy proof (1.5 TB).
- **Opportunity**: Ready for regulated pilots (CBDC, defense quantum-hardening, IoT critical) with measurable advantage over legacy AES-only stacks.

2. Tangible Proof Points

Pillar: Metric; Evidence

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Statistical Integrity: PractRand 1.5 TB (-tf 2) + PQC quick 64 GB;
`practrand\_logs/practrand\_whitened\_20251124\_011640.log`,
`practrand\_logs/pqc\_practrand\_quick\_20251124\_110930.log`

Key Sensitivity: 0.500 bit-flip ratio (512–2048 bit); `reports/key\_sensitivity\_20251124T103534Z.json`

PQC Throughput: Kyber512 avg 0.020 ms; Dilithium3 avg 0.105 ms;
`reports/pqc\_benchmark\_full\_20251124T110127Z.json`

Governance: MPC-N event `diagnostics.pqc\_summary` + diagnostics guard;
`docs/diagnostics/DIAGNOSTICS\_MPCN\_GUARD\_2025-11-23.md`

3. Business Impact

- **Government & CBDC**: PractRand scale + Kyber integration meets tender prerequisites for national payment rails.
- **Defense & Space**: Fishbone multi-layer design gives 47.8% avalanche + 100% reversibility, aligning with zero-loss doctrine.
- **Enterprise SaaS & IoT**: Sub-millisecond PQC performance enables retrofit without hardware swaps; licensing upsell via MPC-N traceability.

4. Competitive Positioning

1. **Proof over Promises**: Only vendor in segment with simultaneous PractRand TB run, PQC-specific PractRand, and liboqs benchmark artifacts.
2. **Audit-Ready**: Documentation cross-linked (report, diagnostics guard, index, MPC-N) reduces due diligence cycle from weeks to hours.
3. **Roadmap Leverage**: QuantumResistanceManager (Rib 4) + CertificationTracker (Rib 6) already specified, providing clear maturity path to 99.5% target.

5. Immediate Calls to Action

1. **Commercial**: Launch investor/partner briefings using this summary + full PQC report for diligence packets.
2. **Technical**: Schedule PractRand PQC Core (512 GB) window to pursue “Selo Máximo PQC”.
3. **Regulatory**: Prepare ISO 27001 / NIST PQC dossier kick-off using existing artifacts and MPC-N logs.

6. Key Contacts & Assets

- **Primary Report**: `docs/PQC\_VALIDATION\_REPORT\_2025-11-24.md`
- **Executive Deck Hooks**: Avalanche data (Section 3.2), PQC throughput (Section 3.3), MPC-N guard (Section 6).
- **Distribution Package**: This summary (PDF), PQC report (PDF/MD), PractRand log excerpts, key sensitivity + benchmark JSONs.

KayosCrypto is now positioned to convert its PQC validation leadership into commercial wins. The remaining gap to 99.5% maturity is strategic execution, not technical feasibility.